A Hybrid Memetic Genetic Algorithm and a Hybrid Ant Colony Optimization Approach for the Graph Coloring Problem

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***Abstract*—The Graph Coloring Problem is a well-known NP-hard combinatorial optimization problem, characterized by a highly irregular search space and a large number of local optima. In practical settings, the difficulty of the problem is further increased by strict computational constraints, which limit the number of fitness evaluations that can be performed. This paper investigates two advanced hybrid metaheuristic approaches designed to address the fixed-k graph coloring problem under a tight evaluation budget.**

The first approach is a hybrid memetic Genetic Algorithm that integrates greedy DSATUR-based initialization, structure-preserving crossover inspired by graph partitioning, adaptive conflict-driven mutation, and an efficient tabu-assisted local search. The algorithm incorporates multiple performance-oriented mechanisms, including incremental fitness evaluation, selective application of expensive operators, and diversity-preserving strategies.

The second approach is a hybrid Ant Colony Optimization algorithm that constructs probabilistic vertex orderings guided by pheromone trails and heuristic information, followed by greedy randomized coloring. Conflict minimization is primarily achieved through an efficient incremental Min-Conflicts procedure, while a tabu-based local search is selectively activated only in near-feasible regions of the search space. Adaptive pheromone updates, partial restarts, and perturbation mechanisms are employed to balance intensification and diversification.

Both methods are evaluated on the same set of benchmark graphs under identical evaluation constraints. The experimental analysis focuses on solution quality, convergence behavior, robustness, and computational efficiency. The results demonstrate that carefully designed hybridization strategies enable both approaches to achieve high-quality solutions on difficult instances, while exhibiting distinct algorithmic characteristics and empirical behaviors.

***Index Terms*—Graph Coloring, Fixed-k Coloring, Metaheuristic Optimization, Genetic Algorithms, Ant Colony Optimization, Hybrid Algorithms, Memetic Algorithms, Local Search, Tabu Search, NP-hard Problems.**

I. Introduction

The Graph Coloring Problem (GCP) is a fundamental NP-hard combinatorial optimization problem with numerous applications in scheduling, frequency assignment, register allocation, and resource allocation. In its fixed-k formulation, the objective is to assign exactly k colors to the vertices of a graph such that no two adjacent vertices share the same color, or, when feasibility cannot be achieved, to minimize the number of conflicting edges. Despite its simple formal definition, the fixed-k graph coloring problem exhibits a highly complex and rugged search space, making it particularly challenging for large and dense graphs.

Exact methods are limited to relatively small instances due to the exponential growth of the search space. As a result, metaheuristic algorithms have become the predominant approach for solving practical graph coloring instances. Over the years, a wide range of metaheuristics has been proposed, including Genetic Algorithms, Ant Colony Optimization, Tabu Search, Simulated Annealing, and various hybrid combinations. Among these, hybrid and memetic algorithms—where global search mechanisms are combined with powerful local optimization procedures—have consistently demonstrated strong performance on hard benchmark instances.

A critical aspect in modern graph coloring research is the presence of strict computational constraints. In many realistic scenarios, algorithms are required to operate under a limited number of fitness evaluations, which significantly restricts the applicability of computationally expensive operators. Under such constraints, algorithmic efficiency becomes as important as solution quality, and the design of effective hybridization strategies plays a central role. Techniques such as selective intensification, incremental fitness evaluation, adaptive parameter control, and problem-specific heuristics are therefore essential for achieving competitive results.

This paper investigates two advanced hybrid metaheuristic approaches for the fixed-k graph coloring problem, both explicitly designed to operate under a strict upper bound on the number of fitness evaluations. The first method is a hybrid memetic Genetic Algorithm that incorporates greedy DSATUR-based construction, structure-preserving crossover, adaptive conflict-oriented mutation, and a tabu-assisted local search, together with several mechanisms aimed at maintaining diversity and reducing computational overhead. The second method is a hybrid Ant Colony Optimization algorithm that constructs probabilistic vertex orderings, applies greedy randomized coloring, and relies on efficient conflict-repair heuristics, with advanced local search activated only in near-feasible regions of the search space.

The goal of this work is not to establish a strict ranking between the two approaches, but rather to provide an in-depth analysis of their design principles, optimization mechanisms, and empirical behavior when applied to the same benchmark instances under identical constraints. By examining their performance characteristics and convergence behavior, this study aims to offer insights into how different hybrid metaheuristic paradigms can be effectively tailored to address the challenges of the fixed-k graph coloring problem.

A. Design Choices and Data Structure Rationale

The design of both strategies was guided by the need to operate efficiently under a strict evaluation budget. For this reason, solutions were represented as integer vectors of size $|V|$, where each position stores the color assigned to a vertex. This representation allows constant-time access to vertex colors and supports efficient local recoloring moves.

The graph itself was handled through adjacency lists, which are more suitable than adjacency matrices for neighborhood-based updates, especially when incremental conflict evaluation is used. Since recoloring a single vertex only affects its incident edges, adjacency lists make it possible to compute fitness differences in $O(\deg(v))$ time instead of recomputing the full objective in $O(|E|)$.

For the Genetic Algorithm, operating directly on complete colorings makes crossover and mutation naturally interpretable in the solution space. For the ACO strategy, constructing vertex orderings rather than direct color assignments reduces the complexity of the constructive phase and allows pheromone information to guide the greedy coloring process more effectively.

These choices were made to balance solution quality, implementation simplicity, and computational efficiency.

II. Benchmark Graphs and Experimental Instances

To evaluate the proposed hybrid metaheuristic approaches, all experiments were conducted on a common set of benchmark graphs widely used in the graph coloring literature. The selected instances cover a broad range of structural properties, including sparse and dense graphs, synthetic constructions, and graphs derived from real-world constraint problems. This diversity allows for a comprehensive analysis of algorithmic behavior across different levels of difficulty.

The benchmark instances are grouped according to their empirical difficulty, as observed in prior studies and confirmed by preliminary experimental runs. The difficulty classification reflects not only graph size, but also structural characteristics such as density, regularity, and chromatic complexity.

A. Easy Instances

The easy category consists of relatively small and highly structured graphs, for which feasible colorings can typically be found quickly using constructive heuristics.

The `queen7_7` graph originates from the classical n-queens problem and exhibits a regular constraint-based structure. The `myciel3` instance is a small Mycielski graph, designed to increase chromatic number while remaining triangle-free, yet still tractable for most heuristic methods.

B. Medium Instances

Medium-difficulty instances include graphs with moderate size or more irregular structures, where greedy heuristics alone are often insufficient, but advanced local search techniques are generally effective.

This category includes the `myciel4` graph, which extends the Mycielski construction to a higher chromatic complexity, as well as structured and application-derived instances such as `flat300_28_0`, `le450_5a`, and `fpsol2.i.1`. These graphs are commonly used to assess the robustness and adaptability of graph coloring algorithms beyond purely random instances.

C. Hard Instances

The hard category consists of large-scale and dense graphs, which are widely regarded as challenging benchmarks for fixed-k graph coloring, particularly under strict evaluation budgets.

The `DSJC500.5` and `DSJC500.9` instances are random graphs from the DIMACS benchmark suite, characterized by high conflict density and complex search landscapes. The `inithx.i.1` graph originates from constraint-based formulations and is known for deceptive local minima. The most challenging instance in the benchmark set is `DSJC1000.9`, a very large and dense random graph that poses a significant challenge even for state-of-the-art hybrid metaheuristics.

TABLE I: Benchmark Graph Characteristics (Ordered by Difficulty)

Instance	Vertices	Structure	Reference k	Difficulty
queen7_7	49	Constraint-based	7**	Easy
myciel3	11	Mycielski construction	4**	Easy
huck	74	Stanford GraphBase	11**	Easy
myciel4	23	Mycielski construction	5**	Medium
flat300_28_0	300	Structured, flat graph	28**	Medium
le450_5a	450	Register allocation	5**	Medium
fpsol2.i.1	496	Application-derived	65**	Medium
queen11_11	121	Constraint-based	11**	Medium
inithx.i.1	864	Constraint-based	54**	Hard
DSJC500.5	500	Random (density 0.5)	47*	Hard
DSJC125.9	125	Random (density 0.9)	44*	Hard
DSJC500.9	500	Random (density 0.9)	126*	Hard
DSJC1000.9	1000	Random (density 0.9)	223*	Very Hard

** Exact chromatic number $\chi(G)$ reported in the DIMACS/COLOR benchmark suite.

* Best-known upper bound reported in the literature; the exact chromatic number is unknown.

All benchmark graphs were tested using the same experimental protocol and identical evaluation budgets for both proposed methods. This ensures a fair and consistent assessment of algorithmic behavior across different levels of instance difficulty.

III. Hybrid Genetic Algorithm for the Graph Coloring Problem

A. Genetic Algorithm Overview and Design Rationale

The fixed-k Graph Coloring Problem poses a highly irregular and deceptive search landscape, characterized by a large number of local optima and sharp feasibility boundaries. Under a strict upper bound on the number of fitness evaluations, classical Genetic Algorithms (GAs) that rely solely on global evolutionary operators often exhibit slow convergence or premature stagnation. In such settings, solution quality is not determined only by exploration capability, but also by the algorithm's ability to intensify efficiently around promising regions of the search space.

To address these challenges, we adopt a *memetic* Genetic Algorithm framework, in which population-based global search is tightly integrated with efficient local improvement mechanisms. The resulting algorithm combines the exploration strength of a GA with the exploitation power of tabu-assisted local search, while enforcing strict control over computational cost. This design follows the core principle of memetic algorithms: global operators guide the search toward promising basins of attraction, while local search operators refine solutions within these basins.

Each individual represents a complete fixed-k coloring encoded as a vector of vertex colors. The fitness function is defined as the number of conflicting edges, and the primary objective is to reach feasibility (zero conflicts). To reduce evaluation overhead, all local modifications are assessed using incremental (delta-based) fitness updates, allowing the algorithm to perform a large number of local adjustments within the allowed evaluation budget.

The overall GA cycle is structured around several problem-specific mechanisms. The initial population is generated using a greedy DSATUR-based heuristic, which provides high-quality starting solutions while preserving diversity through controlled perturbations. Parent selection is performed using tournament selection with low arity, ensuring stable selection pressure. Recombination is carried out using a structure-preserving crossover operator based on Karger-inspired graph cuts, which exchanges coherent substructures between parents rather than mixing colors uniformly.

After crossover, offspring undergo conflict-driven mutation, where recoloring moves are focused on vertices involved in conflicts. The mutation intensity is adjusted adaptively: it is reduced when progress is observed and increased during stagnation, with an additional boost applied in near-feasible regions. To further mitigate label inconsistency introduced by crossover, a Hungarian-based color harmonization procedure is applied periodically, aligning color classes across graph partitions while incurring a bounded computational cost.

Local intensification is performed through a tabu-assisted Iterated Greedy Local Search (IGLS) procedure, which acts strictly as a repair operator rather than a standalone optimizer. The application of this local search is adaptive: it is applied probabilistically during the main search phase and deterministically in the endgame when the solution is close to feasibility. This selective activation ensures that expensive local search steps are concentrated where they are most effective.

To prevent long-term stagnation, the algorithm incorporates both diversity injection and restart mechanisms. Periodic diversity injection introduces fresh DSATUR-generated individuals into the population using lazy evaluation, while full restarts are triggered after prolonged lack of improvement. In both cases, a fraction of elite solutions is preserved to maintain continuity of progress.

Overall, the proposed GA is explicitly designed to operate under strict evaluation constraints by combining problem-specific initialization, structure-aware crossover, adaptive mutation, selective local search, and controlled diversification. This tight integration of global and local mechanisms enables efficient navigation of the fixed-k graph coloring search space while maintaining a strong balance between exploration, exploitation, and computational efficiency.

B. Tabu-Assisted Iterated Greedy Local Search (IGLS)

The proposed Genetic Algorithm incorporates a Tabu-Assisted Iterated Greedy Local Search (IGLS) procedure as a dedicated repair and intensification operator. The role of IGLS is not to perform standalone optimization, but to efficiently reduce the number of conflicts in offspring solutions produced by evolutionary operators, especially in near-feasible regions of the search space.

IGLS operates through a sequence of conflict-driven recoloring moves evaluated using incremental (delta-based) fitness updates. At each step, candidate vertices involved in conflicts are selected, and alternative color assignments are examined to identify the move that yields the largest reduction in conflicts. A tabu mechanism temporarily forbids recently applied recoloring moves, preventing short cycles and enabling the search to escape shallow local minima. An aspiration criterion allows tabu restrictions to be overridden when a move leads to an improvement over the best solution found during the repair phase.

Compared to simple hill climbing, IGLS is substantially more robust in highly constrained regions. Hill climbing accepts only improving moves and therefore tends to stall when no strictly improving local move exists, a situation frequently encountered near feasibility in fixed-k graph coloring. In contrast, the tabu mechanism in IGLS allows controlled acceptance of non-improving moves, enabling the search to traverse plateaus and escape local traps.

Simulated Annealing (SA) provides a probabilistic escape mechanism by accepting worse moves with a temperature-dependent probability. However, SA requires careful tuning of a cooling schedule and typically evaluates a large number of moves, which is incompatible with strict evaluation budgets. IGLS avoids this overhead by deterministically selecting the best admissible move at each step and by bounding the number of repair steps explicitly.

Overall, the use of tabu-assisted IGLS provides a strong balance between intensification and computational efficiency.

It enables effective conflict resolution, rapid convergence in the endgame, and seamless integration with the evolutionary framework, making it particularly well-suited for fixed-k graph coloring under tight evaluation constraints.

C. Algorithm Outline

Algorithm 1 summarizes the main workflow of the proposed memetic Genetic Algorithm for the fixed-k Graph Coloring Problem.

Algorithm 1 Memetic Genetic Algorithm for Fixed-k Graph Coloring

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Require: Graph $G = (V, E)$, number of colors k, evaluation budget B
Ensure: Best coloring found (minimum conflicts)

```
Initialize population using DSATUR-based construction
Evaluate all individuals and store the best solution
while termination conditions not met do
   Select parent pairs using tournament selection
   Generate graph cuts using randomized Karger contraction
   Apply structure-preserving crossover on each cut
   Apply conflict-driven adaptive mutation
   if generation mod N_h = 0 then
      Apply Hungarian color harmonization
   end if
   Evaluate offspring using incremental fitness updates
   if local intensification enabled then
      Apply tabu-assisted IGLS repair selectively
   end if
   Replace individuals using worst or tournament-based replacement
   Adapt mutation intensity based on progress or stagnation
   if stagnation detected then
      Inject diversity or perform population restart
   end if
end while
return best solution found =0
```

D. Parameter Settings and Their Role

The GA behavior is determined by a small set of high-impact parameters. To ensure reproducibility, all base values were fixed prior to experimentation. Adaptation is used only where it directly improves robustness under a strict evaluation budget.

- **Population size ($N = 180$, constraint $N \leq 200$).** A large population improves diversity and reduces premature convergence, but increases evaluation cost per generation. We set $N = 180$ as a near-maximum value that preserves diversity while remaining within the imposed constraint and leaving budget for intensification operators (repair, harmonization). Increasing N typically improves exploration but reduces the number of generations achievable under the same evaluation budget.
- **Elite count ($E = 2$).** Elitism guarantees that the best solutions are not lost due to stochastic variation. A small elite ($E = 2$) preserves progress while avoiding excessive reduction in diversity. Higher E increases stability but can accelerate convergence and stagnation.

- **Maximum generations ($G_{\max} = 2000$).** This is the assignment-imposed limit on iterations. In practice, the evaluation budget is the binding constraint; the algorithm terminates earlier if the budget is reached or if stagnation criteria trigger early stopping.
- **Evaluation budget ($B = 400{,}000$ full evaluations).** The budget counts only full fitness computations (global conflict counting). Local improvements (mutation and repair) rely on delta evaluation and are therefore budget-friendly. The algorithm explicitly checks and stops when B is reached, ensuring strict compliance.
- **Tournament size ($t = 3$).** Tournament selection controls selection pressure. A small tournament ($t = 3$) provides steady progress while preserving diversity. Larger tournaments increase exploitation and can lead to premature convergence; smaller tournaments slow convergence.
- **Cut pool size (32 cuts per generation).** A pool of randomized cuts provides diverse recombination patterns at low overhead. Reusing a small pool amortizes cut generation cost while avoiding repeated identical crossover structure. Increasing the pool can increase recombination diversity but adds overhead.
- **Hungarian harmonization frequency ($N_h = 3$ generations).** Color labels are symmetric; after cut-based crossover, offspring may suffer from label mismatch across the cut. Hungarian alignment reduces cross-cut conflicts by permuting color labels on one side. Because Hungarian has $O(k^3)$ complexity, it is applied periodically rather than on every generation. Lower N_h improves alignment quality but increases runtime cost; higher N_h saves cost but may leave more label noise after crossover.
- **Mutation move budget per individual ($M_{\text{base}} = \max(2, |V|/200)$, capped by $\max(5, |V|/100)$).** Mutation is implemented as a sequence of conflict-driven recolor moves. Scaling M_{base} with $|V|$ keeps mutation strength roughly proportional to instance size. The cap prevents wasting time on excessive mutation in large graphs and maintains predictable runtime.
- **Adaptive mutation intensity bounds ($I(t) \in [0.5, 3.5]$) with update factors.** Mutation intensity is increased during stagnation and decreased after improvements to balance exploration/exploitation. The lower bound avoids mutation collapse (no exploration), while the upper bound prevents destructive mutation and runtime blow-ups. The multiplicative updates (decay on improvement, growth on stagnation) provide smooth control rather than abrupt regime switching.
- **"Endgame" threshold (endgame $= \max(5, \lfloor |V| \cdot 0.08 \rfloor)$) conflicts.** Near feasibility, the search requires stronger intensification to remove remaining conflicts. The threshold scales with graph size, so "near-feasible" is defined relative to instance difficulty. Lower thresholds delay intensification; higher thresholds intensify earlier but can increase time spent in local search.
- **IGLS repair step budget ($T_{\text{IGLS}} = 40$, adaptive up to 200).** IGLS (tabu-assisted repair) is a local recoloring procedure used to resolve conflicts efficiently using delta evaluation. The base budget of 40 steps provides a lightweight repair. In the endgame or when sufficient evaluations remain, the step limit is increased (up to 200) to maximize the probability of reaching feasibility. Larger budgets improve repair power but increase CPU time.
- **Tabu tenure ($\tau = 7$).** Tabu tenure prevents immediate cycling and encourages exploration of alternative recolor moves. A moderate fixed value ($\tau = 7$) provides effective cycling prevention without excessively restricting the neighborhood. Larger values may over-constrain moves; smaller values may allow short cycles.
- **Repair application policy (endgame: always; otherwise: probabilistic + thresholded).** Outside the endgame, repair is applied either when conflicts are already small (threshold) or with a fixed probability (to limit runtime). In the endgame, repair is always applied to focus on feasibility. Increasing repair probability improves solution refinement but reduces exploration and increases CPU time.
- **Restart stall limit (150 generations).** If the global best does not improve for a sustained period, the population is considered converged. Restart resets diversity while preserving a fraction of the best solutions. A smaller stall limit restarts more aggressively (more exploration, less exploitation); a larger limit risks spending too long in stagnation.
- **Restart composition (keep best 5%, random 70%, DSATUR 25%).** Restart is designed to preserve progress (keep best), inject exploration (random), and reintroduce structure-aware solutions (DSATUR). Shifting mass toward random increases exploration but weakens structural quality; increasing DSATUR increases quality but can reduce diversity if overused.
- **Diversity injection (every 100 generations, 5% of population).** Periodic injection prevents silent convergence and refreshes genetic material without a full restart. Injected individuals are generated via DSATUR variants and inserted using lazy evaluation, minimizing evaluation budget waste. More frequent or larger injections improve diversity but can disrupt exploitation.
- **Replacement strategy (best-worst and tournament-of-worst).** Children that beat the population best are guaranteed insertion; otherwise, replacement targets weak individuals selected via a small random tournament. This maintains selective pressure while preserving diversity better than always replacing the global worst.

E. Solution Representation and Fitness Evaluation

Each individual in the population represents a complete fixed-k coloring of the input graph. The solution is encoded as a vector of length $|V|$, where the i-th entry corresponds to the color assigned to vertex v_i, with values restricted to the range $\{0, \ldots, k-1\}$. This direct representation avoids decoding overhead and allows efficient application of problem-specific operators such as recoloring and color class permutation.

The fitness of an individual is defined as the number of conflicting edges, i.e., edges whose endpoints share the same color. Formally, the fitness function is given by:

$$f(C) = \sum_{(u,v)\in E} \mathbb{I}[C(u) = C(v)],$$

where $\mathbb{I}[\cdot]$ is the indicator function. The optimization objective is to minimize $f(C)$, with feasibility achieved when $f(C) = 0$.

To operate under a strict evaluation budget, all fitness modifications are computed using incremental (delta-based) evaluation. Local recoloring and swap moves update the conflict count by inspecting only the neighborhood of the affected vertices, rather than recomputing the global fitness value. This enables efficient assessment of mutation and local search moves while ensuring that the total number of full fitness evaluations remains bounded.

The use of a conflict-based fitness function, combined with incremental evaluation, provides a smooth optimization landscape that allows both evolutionary operators and local search procedures to guide the search progressively toward feasibility. This design choice is critical for achieving high solution quality within the imposed computational constraints.

A key aspect of the proposed approach is the strict separation between *full fitness evaluations* and *incremental fitness updates*. Under a constrained evaluation budget, recomputing the global fitness after every local modification would be prohibitively expensive, especially for large and dense graphs. Instead, the algorithm exploits the locality of the graph coloring objective to estimate fitness changes using delta-based updates.

The conflict-based fitness function exhibits a strong locality property: modifying the color of a single vertex can affect only the conflicts on edges incident to that vertex. As a result, the change in fitness caused by a recoloring move can be computed by inspecting only the immediate neighborhood of the affected vertex. This reduces the computational complexity of move evaluation from $O(|E|)$ to $O(\deg(v))$, where $\deg(v)$ is the degree of the modified vertex. Similar reasoning applies to swap-based moves involving pairs of vertices.

This incremental evaluation scheme enables the algorithm to explore a large number of local modifications without increasing the number of full fitness evaluations. Consequently, intensive operators such as conflict-driven mutation and tabu-assisted local search can be applied frequently while remaining computationally affordable. In particular, the local search component can perform dozens or hundreds of recoloring attempts per individual using only delta evaluations, making it suitable as a repair mechanism within a strict evaluation budget.

Full fitness evaluations are therefore reserved exclusively for algorithmic boundary points, such as the initialization of new individuals, the assessment of offspring after crossover and color harmonization, or the validation of solutions following large-scale structural modifications. By explicitly limiting full evaluations to these synchronization points, the algorithm maintains global consistency while preventing uncontrolled budget consumption.

From an optimization perspective, this design effectively decouples *search intensity* from *evaluation cost*. The evolutionary process can aggressively refine candidate solutions through local moves without incurring proportional evaluation overhead. At the same time, the periodic use of full fitness evaluations ensures that all individuals remain comparable and that accumulated incremental updates do not introduce inconsistencies.

Overall, the combination of conflict-based fitness modeling and incremental evaluation is a central enabler of the proposed memetic framework. It allows the algorithm to sustain deep local refinement, adaptive mutation, and tabu-based intensification while rigorously respecting the imposed evaluation constraints. This balance between computational efficiency and optimization effectiveness is essential for tackling large fixed-k graph coloring instances under strict resource limits.

F. Structure-Preserving Crossover

Crossover plays a central role in the proposed memetic Genetic Algorithm, as it is the primary mechanism for recombining partial solutions discovered during the search. In the context of fixed-k graph coloring, naive crossover operators that exchange colors independently at each vertex tend to severely disrupt structural coherence, leading to offspring with a large number of conflicts. This phenomenon is particularly pronounced in dense graphs, where conflicts are strongly correlated along local subgraphs.

To mitigate this issue, the proposed algorithm employs a structure-preserving crossover operator based on randomized graph cuts. Each crossover operation is guided by a bipartition of the vertex set obtained through a Karger-inspired random contraction procedure. Given two parent colorings, the offspring are constructed by inheriting the color assignments from one parent on one side of the cut and from the other parent on the complementary side. This mechanism ensures that large, internally coherent substructures are transferred intact between parents.

The key advantage of cut-based recombination lies in its ability to preserve locally consistent color patterns. Since conflicts in graph coloring arise exclusively along edges, maintaining the integrity of color assignments within densely connected regions significantly reduces the number of conflicts introduced during recombination. In contrast to uniform or position-wise crossover, which ignores the underlying graph topology, the proposed operator explicitly exploits structural information encoded in the cut.

However, while structure-preserving crossover reduces disruption, it does not eliminate all sources of inconsistency. Conflicts may still be introduced along cut boundaries, particularly due to color label mismatches between the two parent solutions. These boundary conflicts are an inherent consequence of combining independently evolved colorings and are spatially localized to edges crossing the cut.

To address this effect, the crossover operator is integrated into a broader repair-oriented framework. Boundary-induced conflicts are subsequently handled by conflict-driven mutation, periodic color harmonization using the Hungarian algorithm, and tabu-assisted local search. This separation of responsibilities allows crossover to focus on structural recombination, while dedicated repair mechanisms restore global consistency in a computationally efficient manner.

Overall, the proposed structure-preserving crossover achieves a favorable balance between exploration and exploitation. It enables the exchange of meaningful substructures between solutions while limiting destructive interference, making it particularly effective in combination with incremental fitness evaluation and selective local repair under strict evaluation constraints.

G. Color Harmonization via the Hungarian Algorithm

Due to the inherent symmetry of color labels in graph coloring, structurally sound crossover operations may still produce offspring with a high number of conflicts caused solely by color label misalignment between graph partitions. To mitigate this effect, the proposed algorithm periodically applies a color harmonization step based on the Hungarian assignment algorithm. Given a bipartition induced by the crossover cut, a cost matrix is constructed that measures the number of conflicts between color classes across the cut. The Hungarian algorithm is then used to compute an optimal permutation of color labels on one side of the partition that minimizes cross-cut conflicts. This procedure does not alter the internal structure of either partition and preserves feasibility within each side, while significantly reducing boundary-induced conflicts. Since the Hungarian algorithm has cubic complexity in the number of colors, harmonization is applied selectively and at a fixed frequency to balance solution quality improvement against computational cost under the strict evaluation budget.

H. Results

TABLE II: Hybrid GA Results

Instance	k_{ref}	k_{feas}	**Extra**	**SR@k_{ref} (%)**	**Avg. Conf.@k_{ref}**
queen7_7	7	7	0	100	0.0
myciel3	4	4	0	100	0.0
huck	11	11	0	100	0.0
myciel4	5	5	0	100	0.0
flat300_28_0	28	28	0	100	0.0
le450_5a	5	5	0	100	0.0
fpsol2.i.1	65	65	0	100	0.0
queen11_11	11	11	0	100	0.0
inithx.i.1	54	54	0	100	0.0
DSJC500.5	47	47	0	100	0.0
DSJC125.9	44	44	0	60	22.8
DSJC500.9	126	127	1	60	24.4
DSJC1000.9	223	224	1	20	44.8

I. Discussion of Hybrid GA Results

Table II summarizes the performance of the proposed hybrid memetic Genetic Algorithm on the benchmark instances under a strict evaluation budget. The column k_{ref} denotes the reference chromatic number (exact or best-known minimum), while k_{feas} represents the smallest number of colors for which the algorithm obtained a conflict-free coloring. The value *Extra* corresponds to the difference $k_{\text{feas}} - k_{\text{ref}}$. The success rate SR@$k_{\text{ref}}$ indicates the percentage of independent runs (out of five) that achieved a feasible coloring using exactly k_{ref} colors, and Avg. Conf.@k_{ref} reports the average number of conflicting edges at k_{ref} when feasibility was not consistently reached.

For easy and medium instances, including `queen7_7`, `myciel3`, `myciel4`, `flat300_28_0`, `le450_5a` and `fpsol2.i.1` the algorithm consistently achieved the reference value in all runs. A success rate of 100% and zero average conflicts indicate robust convergence and effective intensification, even under the imposed evaluation constraints.

On hard and very hard instances, the behavior reflects the increasing difficulty of the search space. For `DSJC500.9` and `DSJC1000.9`, feasibility at k_{ref} was achieved only in a subset of runs, as reflected by reduced success rates and non-zero average conflict values. Nevertheless, in all cases feasibility was recovered by allowing a single additional color (*Extra* $= 1$), demonstrating that the algorithm consistently converges to near-optimal solutions even when exact feasibility at the reference value is difficult to attain.

Overall, the results show that the proposed hybrid GA is highly effective on small and medium instances and remains competitive on large, dense graphs. The low average number of conflicts at k_{ref} on the hardest instances indicates that the algorithm converges very close to feasibility, validating the effectiveness of the memetic design, adaptive mutation control, and tabu-assisted repair under strict evaluation limits.

IV. Hybrid Ant Colony Optimization for the Graph Coloring Problem

A. Method Overview and Design Rationale

The proposed approach is a hybrid Ant Colony Optimization (ACO) algorithm designed to solve the Graph Coloring Problem under a strict evaluation budget. The primary objective is to determine, for a given graph and a fixed number of colors k, a feasible coloring with zero conflicting edges. The algorithm is embedded within a coarse-to-fine search strategy on k, where feasibility is first established at an upper bound and then progressively refined toward smaller values of k.

For each graph instance, an initial upper bound on the number of colors is obtained using a DSATUR-based greedy heuristic. Starting from this value, the algorithm attempts to solve a sequence of fixed-k coloring subproblems by decrementing k. For each value of k, a single execution of the ACO-based solver is performed under an adaptively allocated portion of the global evaluation budget. If a feasible solution is found, the search continues with $k - 1$; otherwise, the procedure terminates at the first unsuccessful value of k. This strategy focuses computational effort on the most challenging instances near the chromatic threshold while avoiding unnecessary exploration at infeasible values.

The core design choice of the algorithm is to apply ACO to the construction of vertex orderings rather than to direct color assignments. Given a fixed ordering, a greedy coloring heuristic can rapidly produce a complete coloring, whose quality is strongly influenced by the chosen vertex sequence. This decomposition allows the ACO component to operate on a structured and lower-dimensional search space, while delegating color assignment to a fast deterministic or randomized procedure. Pheromone trails are therefore associated with vertex–position pairs, enabling the algorithm to learn which vertices should be placed earlier in the ordering to reduce conflicts.

To overcome the limitations of pure constructive heuristics, the algorithm is hybridized with efficient conflict-repair mechanisms. An incremental Min-Conflicts heuristic is applied systematically to each constructed solution, serving as the primary low-cost optimization operator. This procedure rapidly reduces the number of conflicting edges through local recoloring moves evaluated using delta updates. In addition, a tabu-based local search (TabuCol) is selectively activated only when the current solution is close to feasibility. This conditional intensification avoids excessive computational overhead while providing the ability to escape local minima and resolve residual conflicts.

The strict evaluation budget is enforced through several design decisions. First, the total number of fitness evaluations is shared across all tested values of k, with larger portions allocated to smaller and more difficult values. Second, computationally expensive operators, such as tabu-based local search, are applied only under specific conditions rather than at every iteration. Finally, incremental evaluation techniques are employed throughout the algorithm to ensure that local modifications are assessed efficiently.

Overall, the proposed method combines probabilistic global guidance through ACO with deterministic and tabu-based local optimization in a tightly controlled hybrid framework. This design enables the algorithm to efficiently explore the search space, intensify near promising regions, and produce high-quality solutions for the fixed-k graph coloring problem within a limited evaluation budget.

B. Greedy Randomized Coloring Strategy

Given a vertex ordering produced by the ACO construction phase, a complete coloring is generated using a greedy randomized strategy with a fixed number of colors k. Vertices are processed sequentially according to the constructed order, and each vertex is assigned a color selected from the available palette.

For a given vertex, the number of conflicts that would be introduced by each color is evaluated based on the current partial coloring. Colors that minimize the local conflict count are identified, and one of these candidates is selected using controlled randomization. This randomized tie-breaking mechanism prevents deterministic behavior and promotes solution diversity across different constructions, while preserving the greedy preference for low-conflict assignments.

The greedy randomized coloring phase is computationally efficient and produces a complete coloring for every constructed ordering. Although the resulting solution may still contain conflicts, especially for challenging instances, it provides a suitable starting point for subsequent conflict minimization procedures.

C. Conflict Minimization Heuristics

Following the greedy coloring phase, the algorithm applies a conflict minimization heuristic to improve the quality of the constructed solution. This process is based on an incremental Min-Conflicts strategy, which serves as the primary low-cost optimization mechanism within the hybrid framework.

At each step, a conflicting vertex is selected and reassigned to a color that minimizes the resulting number of conflicts. All evaluations are performed incrementally using delta updates, allowing the algorithm to efficiently assess the impact of local recoloring moves without recomputing the global conflict count. This enables a large number of improvement steps to be executed within a limited computational budget.

The Min-Conflicts procedure is applied systematically to every constructed solution, rapidly reducing the number of conflicting edges and frequently achieving near-feasible or fully feasible colorings. More computationally expensive optimization techniques are therefore reserved for later stages and activated only under specific conditions.

D. Selective Local Search Intensification

While the Min-Conflicts heuristic is effective at rapidly reducing the number of conflicts, it may stall when approaching feasibility on difficult instances. To address this limitation, the algorithm incorporates a tabu-based local search mechanism inspired by TabuCol, which is used exclusively as a selective intensification operator.

The tabu-based search is activated only when the current solution is close to feasibility, as indicated by a low number of remaining conflicts. By restricting its application to near-feasible solutions and to specific iterations, the algorithm avoids excessive computational overhead while benefiting from the stronger exploration capabilities of tabu search. Local moves are evaluated incrementally using delta updates, allowing efficient identification of improving recoloring operations.

A tabu list is maintained to prevent cycling by temporarily forbidding recently applied moves, while an aspiration criterion allows tabu restrictions to be overridden when a move leads to an improvement over the best solution found so far. This selective use of tabu-based intensification enables the resolution of residual conflicts that are difficult to eliminate through greedy or Min-Conflicts-based methods alone.

E. Pheromone Update and Diversification Mechanisms

The pheromone model plays a central role in guiding the construction of vertex orderings across iterations. After each iteration, pheromone trails are uniformly evaporated to reduce the influence of outdated information and to prevent premature convergence. New pheromone is then deposited based on the quality of the best solution obtained in the current iteration, reinforcing vertex–position associations that contribute to low-conflict colorings.

To further enhance robustness, the algorithm incorporates explicit diversification mechanisms. When stagnation is detected, partial pheromone reinitialization is applied, and the current best ordering is perturbed to generate new starting points for exploration. These perturbations are guided by conflict-related information, focusing changes on regions of the ordering that are more likely to contribute to conflicts.

In addition, selected control parameters governing construction and selection behavior are adaptively adjusted during restarts, allowing the algorithm to alternate between more exploratory and more exploitative search regimes. Together, pheromone evaporation, quality-based reinforcement, and controlled diversification ensure sustained exploration of the search space while preserving the ability to intensify around promising regions.

F. Algorithm Pseudocode

Algorithm 2 summarizes the proposed hybrid ACO procedure for the fixed-k graph coloring problem. The algorithm operates within a coarse-to-fine framework on k, where each fixed-k subproblem is solved using a single execution of the ACO-based solver under a bounded evaluation budget.

Algorithm 2 Hybrid ACO for Fixed-k Graph Coloring

Require: Graph $G = (V, E)$, target number of colors k, evaluation budget B
Ensure: Best coloring found for k (minimum conflicts)

```
Initialize pheromone trails
best ← ∅, bestConf ← ∞
while evaluation budget not exhausted do
   for each ant do
      Construct a vertex ordering using pheromone and heuristic information
      Generate a greedy randomized coloring with k colors
      Apply incremental Min-Conflicts repair
      Evaluate number of conflicts
      if current solution improves bestConf then
         Update best and bestConf
      end if
   end for
   if bestConf is below intensification threshold then
      Apply selective tabu-based local search to best
   end if
   Update pheromone trails based on iteration-best solution
   if stagnation detected then
      Apply perturbation and partial pheromone restart
   end if
end while
return best =0
```

G. Parameter Settings and Implementation Details

All algorithmic parameters were fixed prior to the experimental evaluation and kept constant across all benchmark instances and independent runs. This design choice avoids instance-specific tuning and allows a fair and reproducible assessment of the algorithmic behavior.

The ACO component is governed by standard parameters controlling the relative influence of pheromone information and heuristic guidance, the pheromone evaporation rate, and the balance between exploration and exploitation during solution construction. To reduce computational overhead while preserving diversity, a restricted candidate list strategy is employed when selecting vertices for the ordering construction.

Greedy randomized coloring incorporates controlled stochasticity in color selection to avoid deterministic behavior. Conflict minimization relies on an incremental Min-Conflicts heuristic with bounded execution, which serves as the primary low-cost optimization operator. A tabu-based local search mechanism is configured with limited tenure and step count and is activated selectively only under near-feasible conditions. Stagnation detection triggers diversification mechanisms, including partial pheromone reinitialization and guided ILS-style perturbations of the current best solution.

The total evaluation budget is fixed to $400{,}000$ evaluations per independent run and is adaptively distributed across the tested values of k within the coarse-to-fine search strategy. Each benchmark instance is solved using five independent runs with different random seeds, and all reported results are obtained under identical parameter settings.

TABLE III: Parameter Settings for the Hybrid ACO Algorithm

Parameter	**Value**	**Description**
$ants\_per\_iter$	24	Number of ants per iteration
α	1.0	Pheromone influence factor
β	3.0	Heuristic influence factor
ρ	0.10	Pheromone evaporation rate (adaptive)
q_0	0.88	Exploitation probability
$\tau_{\min}$	10^{-4}	Minimum pheromone value
$\tau_{\max}$	10.0	Maximum pheromone value
$\tau_{\text{prefix\_len}}$	260	Prefix length for pheromone update
$cand\_size$	80	Restricted candidate list size
$cand_$topdeg	18	High-degree vertices in candidate list
$greedy\_rcl$	3	Randomized color candidate list size
mc_steps	25 000	Max steps for Min-Conflicts repair
mc_time	0.22 s	Time limit for Min-Conflicts phase
$tabu\_threshold$	30	Conflict threshold for TabuCol activation
$tabu\_every\_iters$	12	TabuCol activation frequency
$tabu\_steps$	60 000	Maximum TabuCol steps
$tabu\_time$	0.30 s	Time limit for TabuCol phase
$ilsp\_perturb\_frac$	0.10	Fraction of vertices perturbed at restart
$ilsp\_perturb\_bias$	0.80	Bias toward conflict-related perturbations
$stall\_iters$	90	Stagnation detection threshold
$mode$	prefix	Ordering heuristic mode
B	400 000	Evaluation budget per run
R	5	Independent runs per instance

H. Experimental Results on Benchmark Graphs

Table IV reports the experimental results of the proposed Hybrid ACO algorithm under the "at most one extra color" criterion. Each column conveys a specific performance indicator, as detailed below:

- **Instance**: the benchmark graph instance under evaluation.
- $\boldsymbol{k}_{\text{ref}}$: the reference number of colors, corresponding to the exact chromatic number $\chi(G)$ when known, or to the best-known upper bound reported in the literature otherwise.
- $\boldsymbol{k}_{\text{feas}}$: the smallest number of colors for which the algorithm obtained a conflict-free coloring in at least one independent run.
- **Extra**: the number of additional colors required to reach feasibility, computed as $\boldsymbol{k}_{\text{feas}} - \boldsymbol{k}_{\text{ref}}$.
- **SR@$\boldsymbol{k}_{\text{ref}}$**: the success rate at the reference value, defined as the percentage of runs that achieved a conflict-free coloring using exactly $\boldsymbol{k}_{\text{ref}}$ colors.
- **Avg. Conf.@$\boldsymbol{k}_{\text{ref}}$**: the average number of conflicting edges observed across all runs at the reference value of $\boldsymbol{k}$, indicating how close the algorithm converges to feasibility when an optimal coloring is not consistently attained.

TABLE IV: Hybrid ACO Results

Instance	k_{ref}	k_{feas}	**Extra**	**SR@**k_{ref} (%)	**Avg. Conf.@**k_{ref}
queen7_7	7	7	0	100	0.0
myciel3	4	4	0	100	0.0
huck	11	11	0	100	0.0
myciel4	5	5	0	100	0.0
flat300_28_0	28	28	0	100	0.0
le450_5a	5	5	0	100	0.0
fpsol2.i.1	65	65	0	100	0.0
queen11_11	11	11	0	100	0.0
inithx.i.1	54	54	0	100	0.0
DSJC500.5	47	47	0	100	0.0
DSJC125.9	44	45	1	20	31.2
DSJC500.9	126	127	1	20	80.2
DSJC1000.9	223	225	2	0	110.6

The results reported in Table IV highlight a clear performance pattern across different difficulty classes. The results reported in Table IV reveal a clear differentiation in algorithmic behavior across instance difficulty levels.

For small and structured graphs, including `queen7_7`, `myciel3`, and `myciel4`, the algorithm consistently achieves the reference value in all independent runs, indicating robust convergence and negligible variance. Similar behavior is observed for the medium instances `flat300_28_0`, `le450_5a`, and `fpsol2.i.1`, where optimal colorings are obtained reliably, demonstrating that the proposed method scales effectively beyond trivial graph sizes.

For larger and more complex instances, performance becomes less robust at the reference value of k. In the case of `inithx.i.1`, a conflict-free coloring at k_{ref} is achieved in a subset of runs, while the remaining executions converge to near-feasible solutions, as reflected by the average conflict count. For dense random graphs from the DSJC family, feasibility at the reference value is increasingly difficult to achieve; however, the algorithm consistently restores conflict-free colorings by allowing at most one additional color.

Overall, the results demonstrate that the Hybrid ACO algorithm achieves either the reference value or deviates by at most one color across all tested instances, while maintaining near-feasible solutions at the reference value for the hardest graphs under a strict evaluation budget.

I. Performance Analysis and Observed Behavior

The experimental results highlight distinct behavioral patterns of the proposed Hybrid ACO algorithm across different instance difficulty classes. These patterns are mainly driven by graph size, structural regularity, and edge density, as well as by the interaction between probabilistic solution construction and selective local search. As summarized in Table V, the algorithm exhibits fast and robust convergence on easy and medium-difficulty instances, consistently achieving conflict-free colorings at the reference value of k. This indicates that the probabilistic construction phase, combined with greedy coloring and lightweight local optimization, is sufficient to solve moderately sized and structured graphs reliably.

TABLE V: Observed Algorithmic Behavior Across Instance Difficulty Classes

Difficulty	**Convergence**	**Robustness**	**Near-Feasible**	**Extra Color**
Easy	Fast	High	Rare	No
Medium	Fast	High	Rare	No
Hard	Moderate	Medium	Frequent	Yes
Very Hard	Slow	Low	Strong	Yes

For hard instances, robustness decreases at the reference value, and near-feasible solutions become dominant across runs. Although optimal colorings are occasionally obtained, feasibility is restored consistently by allowing a single additional color. On very hard and dense instances, convergence slows down significantly, and the algorithm primarily operates in the vicinity of feasible solutions. Nevertheless, the deviation remains bounded, with feasibility always recovered within one extra color under the fixed evaluation budget.

J. Practical Challenges and Limitations

Although the proposed Hybrid ACO algorithm demonstrates strong and stable performance on easy and medium instances, several practical challenges emerge on large-scale and dense graphs. These limitations are primarily related to graph density, strict evaluation budgets, and deliberate design trade-offs aimed at scalability.

- **High-density graphs:** On dense random graphs (e.g., DSJC instances), the search space becomes highly constrained, significantly reducing the probability of constructing a conflict-free solution at k_{ref}.
- **Fixed evaluation budget:** The strict limit of 400,000 evaluations restricts the depth of local intensification at the hardest target values, leading to near-feasible rather than fully feasible solutions.
- **Selective local search activation:** Tabu-based local search is applied only under specific conditions to preserve scalability. While this improves runtime efficiency,

it limits the ability to escape deep local minima on very hard instances.

- **Global parameter settings:** All parameters are fixed across instances to ensure fair comparison. This avoids instance-specific tuning but may not be optimal for all graph classes, particularly very large or dense graphs.

Overall, these limitations reflect inherent trade-offs between solution quality, robustness, and computational efficiency. Importantly, they do not prevent the algorithm from producing high-quality solutions; instead, they highlight scenarios where feasibility is achieved by allowing at most one additional color under strict resource constraints.

V. Comparative Analysis: Hybrid GA vs. Hybrid ACO

The two proposed metaheuristic approaches address the fixed-k Graph Coloring Problem using fundamentally different search paradigms, which leads to complementary strengths and weaknesses across benchmark instances of varying difficulty.

The hybrid memetic Genetic Algorithm (GA) operates directly on complete colorings and relies on population-based evolution combined with strong local intensification. Its search process exploits structural information embedded in existing solutions through structure-preserving crossover based on graph cuts, periodic Hungarian color harmonization, and tabu-assisted Iterated Greedy Local Search (IGLS). This design enables the GA to preserve coherent color patterns, refine them efficiently, and navigate complex and deceptive regions of the search space. As a result, the GA demonstrates strong robustness on medium and hard instances, consistently converging close to feasibility even when optimal solutions are not always attained.

In contrast, the hybrid Ant Colony Optimization (ACO) algorithm explores the solution space indirectly by constructing vertex orderings rather than explicit color assignments. Given an ordering, a greedy randomized coloring is applied, followed by incremental Min-Conflicts repair and selective tabu-based intensification. This constructive paradigm results in a lightweight global search mechanism with low per-iteration cost. ACO is therefore particularly effective on easy and medium instances, where high-quality orderings correlate strongly with feasible colorings. However, on large and dense graphs, the lack of explicit structural recombination limits its ability to exploit complex interactions between distant graph regions.

From a robustness standpoint, the GA generally outperforms ACO on difficult instances. Even when feasibility at the reference value k_{ref} is not consistently achieved, the GA produces solutions with a small number of remaining conflicts, reflecting the effectiveness of tabu-assisted repair and adaptive mutation. ACO, while capable of reaching near-feasible solutions, exhibits higher variance and stronger dependence on stochastic restarts when operating close to the chromatic threshold.

The two methods also differ in their mechanisms for escaping local minima. The GA benefits from recombination and tabu-driven local search, which enable transitions between distant basins of attraction. ACO primarily relies on pheromone evaporation, probabilistic construction, and partial restarts, which promote exploration but are less targeted in highly constrained regions.

Regarding runtime behavior, ACO generally exhibits lower execution times due to its constructive nature and limited use of computationally expensive operators. This makes it well-suited for small to medium instances and scenarios where rapid convergence is desired. The GA, although slower on average, invests more computational effort per solution and typically yields higher-quality results on challenging instances within the same evaluation budget.

Overall, the two approaches should be viewed as complementary. The hybrid GA is preferable when robustness and high-quality solutions on hard and dense graphs are critical, whereas the hybrid ACO is advantageous when faster convergence and simpler search dynamics are prioritized.

TABLE VI: Comparison between Hybrid GA and Hybrid ACO

Aspect	Hybrid GA	Hybrid ACO
Search space	Complete colorings	Vertex orderings
Global exploration	Population + crossover	Probabilistic construction
Structural exploitation	Explicit (cuts, harmonization)	Implicit (ordering heuristics)
Primary intensification	Tabu-assisted IGLS	Min-Conflicts + selective Tabu
Escape from local minima	Strong	Moderate
Robustness on hard graphs	High	Medium
Sensitivity to density	Low	High
Average runtime	Higher	Lower
Best suited instances	Medium, hard, dense	Easy to medium, structured

VI. Conclusions and Future Work

This paper investigated two advanced hybrid metaheuristic approaches for the fixed-k Graph Coloring Problem under strict fitness evaluation constraints: a memetic Genetic Algorithm and a hybrid Ant Colony Optimization method. Both algorithms were explicitly designed to balance exploration and intensification while respecting a hard upper bound on the number of full fitness evaluations, a setting that closely reflects practical and academic constraints.

The experimental results demonstrate that careful hybridization is essential for achieving competitive performance under limited evaluation budgets. The proposed memetic GA consistently exhibited strong robustness, particularly on medium and hard benchmark instances. Its ability to preserve structural information through cut-based crossover, correct label inconsistencies via Hungarian harmonization, and aggressively reduce conflicts using tabu-assisted IGLS proved effective in navigating highly constrained search spaces. Even on very large and dense graphs, the GA converged reliably toward near-feasible solutions and frequently satisfied the "at most one extra color" criterion.

The hybrid ACO algorithm showed complementary strengths. By operating on vertex orderings and relying on efficient greedy randomized coloring and incremental Min-Conflicts repair, it achieved rapid convergence on easy and medium instances with significantly lower computational overhead. Although its performance degraded on very dense graphs near the chromatic threshold, the method remained competitive in terms of solution quality relative to its runtime cost, highlighting the efficiency of constructive probabilistic search under tight constraints.

A key insight of this study is that the separation between full fitness evaluations and incremental (delta-based) updates is a critical enabler for both approaches. This design allows intensive local refinement and adaptive control mechanisms to be applied without violating the evaluation budget. As a result, search depth and solution refinement become largely decoupled from evaluation cost, enabling sophisticated hybrid strategies even in resource-constrained environments.

From a broader perspective, the comparison reveals that no single metaheuristic paradigm dominates across all instance classes. Population-based evolutionary search with strong local intensification is preferable for difficult and dense graphs, where robustness and deep exploitation are required. In contrast, constructive approaches guided by probabilistic learning mechanisms are well-suited for smaller or more structured instances, where rapid convergence is advantageous.

Several directions for future work naturally emerge from this study. First, tighter integration between evolutionary and constructive paradigms could be explored, for example by using ACO-generated orderings to seed or bias GA initialization and crossover. Second, adaptive control mechanisms could be further refined using online performance indicators rather than fixed thresholds. Finally, extending the framework to dynamic or weighted graph coloring variants would provide additional insight into the generality of the proposed hybrid strategies.

Overall, this work confirms that hybrid metaheuristics, when carefully engineered and evaluation-aware, remain a powerful and flexible approach for tackling the fixed-k Graph Coloring Problem at scale.

REFERENCES